

Nathan Robinson

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Mechanical Engineering student with experience in CAD, manufacturing, controls, and robotics.

EDUCATION

University of California, Los Angeles, BS Mechanical Engineering Sep 2023 – Jun 2027

- GPA: 3.46/4.0
- Relevant Coursework: Dynamical Systems, Controls, Mechanisms, Manufacturing, Strength of Materials, Fluid Mechanics, CAD, C++ , MATLAB, Thermodynamics

ENGINEERING EXPERIENCE

Ergonomics Project Engineer, UCLA Baja SAE Sep 2024 – Jun 2026

- Developed a LiDAR-to-CAD workflow for driver models by reducing initial LiDAR mesh complexity in MeshLab, segmenting limb geometry in Fusion 360, and assembling jointed bodies in SolidWorks using ball joints
- Created multiple driver-specific models to inform cockpit packaging and chassis design, ensuring rules compliance while increasing viable driver range by 15%
- Designed and fabricated an adjustable ergonomic jig to validate steering wheel and pedal placement from SolidWorks models
- Manufactured lightweight carbon fiber seat and steering wheel using wet layup, ensuring rigidity for structural reliability

Undergraduate Researcher — Mechatronics Hardware, UCLA MACLAB Jun 2026 – Present

- Designing and manufacturing calibration jigs and mechanisms for surgical robotics and drone systems
- Contributing to shared GitHub repositories for design documentation and hardware integration

PROJECTS

MATLAB Robot Simulation Feb 2026 – Mar 2026

- Developed a 3-DOF robotic arm simulation in MATLAB using inverse kinematics
- Designed the robot in SolidWorks and integrated the model into Simulink and Simscape Multibody
- Implemented end-effector position control by converting user-specified coordinates into joint motions

Rocket Project Internal Structures Oct 2025 – Mar 2026

- Designed, analyzed, and manually machined an aluminum thrust bulkhead for a student rocket, validating fastener selection and bearing stresses through hand calculations and improving load-bearing capacity by 12

Ergonomic Adjustable Leg Rest Jul 2025 – Aug 2025

[Documentation]

- Modeled and fabricated adjustable-height drawers to improve user posture and leg comfort
- Validated structural design for loads up to 150 lbf through FEA in SolidWorks simulation

Faraday Cage Bed Dec 2025 – Mar 2026

[Documentation]

- Designed and fabricated a bed frame integrating non-permanent fixtures, hinged access, EMF-blocking ventilation, and a programmable LED wake-up system
- Performed structural analysis using ANSYS to validate strength and stiffness assumptions
- Ensured continuous conductivity to maintain effective electromagnetic shielding

WORK EXPERIENCE

E-Commerce Associate, Whole Foods Market Jul 2025 – Sep 2025

- Maintained an average of 90 units/hour, exceeding performance targets by 10%
- Received return offer based on performance and consistency

SKILLS

Design & Analysis: SolidWorks (CAD, Drawings, Simulation, CAM), ANSYS, Fusion 360 (CAD, CAM)

Manufacturing & Fabrication: CNC Milling, Manual Milling, Lathe, Carbon Fiber Layup, 3D Printing, Laser Cutting, Welding

Programming: MATLAB, Simulink, Simscape Multibody, Python, C++ , Arduino, ESP32